

Raaghav Ramani

CONTACT INFORMATION

Center for Nonlinear Studies
Los Alamos National Laboratory
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EMPLOYMENT

Los Alamos National Laboratory

2024–present

Mark Kac Postdoctoral Fellow in Applied Mathematics
Center for Nonlinear Studies

EDUCATION

University of California, Davis

2018–2023

Ph.D. in Applied Mathematics

Advisor: Prof. Steve Shkoller

Dissertation: *PDE-based methods for multiscale gas dynamics simulations*

Trinity College, University of Oxford

2012–2016

MMath Masters in Mathematics, Double First Class Honours

Advisor: Prof. Gregory Seregin

Thesis: *Regularity theory for the axially symmetric incompressible Navier-Stokes equations without swirl*

RESEARCH INTERESTS

PDEs, fluid mechanics, computational physics.

Specifically: wave propagation, moving interface and free boundary problems, multiscale modeling, high-resolution WENO schemes, artificial viscosity, adaptive mesh generation, semi-Lagrangian and Arbitrary Lagrangian-Eulerian schemes, vortex methods, spectral methods, fluid instabilities, combustion, magnetohydrodynamics, computational shock formation and development, topological transitions.

PUBLICATIONS

R. RAMANI

Interface tracking with Microscale Topological Surgery for two-dimensional filament breakup, submitted (2026)

R. RAMANI

A boundary integral method for wave scattering in a heterogeneous medium with a moving obstacle, submitted (2026)

R. RAMANI & S. SHKOLLER

Computational shock development: accurately resolving cusps and discontinuities in solutions of 1D Euler, submitted (2025)

R. RAMANI & S. SHKOLLER

A fast dynamic smooth adaptive meshing scheme with applications to compressible flow
Journal of Computational Physics, 490:112280 (2023)

R. RAMANI & S. SHKOLLER
A multiscale model for Rayleigh-Taylor and Richtmyer-Meshkov instabilities
 Journal of Computational Physics, 405:109177 (2020)

R. RAMANI, J. REISNER, & S. SHKOLLER
A space-time smooth artificial viscosity method with wavelet noise indicator and shock collision scheme, Part 2: The 2-D case
 Journal of Computational Physics, 387:45 – 80 (2019)

R. RAMANI, J. REISNER, & S. SHKOLLER
A space-time smooth artificial viscosity method with wavelet noise indicator and shock collision scheme, Part 1: The 1-D case
 Journal of Computational Physics, 387:81 – 116 (2019)

FELLOWSHIPS & AWARDS	<i>Mark Kac Postdoctoral Fellowship in Applied Mathematics</i>	2024–2026
	Los Alamos National Laboratory	
	<i>Yueh-Jing Lin Scholarship</i>	2023
	Department of Mathematics, University of California, Davis	
	<i>CSES Student Fellowship</i>	2021–2023
	Center for Space & Earth Science, Los Alamos National Laboratory	
	<i>Graduate Research Fellowship</i>	2019–2023
	Department of Mathematics, University of California, Davis	
	<i>Research Fellowship</i>	2016–2018
	Department of Mathematics, University of California, Davis	
	<i>Met Office Masters Thesis Prize</i>	2016
	University of Oxford	
	<i>Mitchell Scholarship</i>	2014
	University of Oxford	
	<i>Scholarship</i>	2013–2016
	Trinity College, University of Oxford	

TALKS	<i>Microscale Topological Surgery for tracking filament breakup</i>	
	CNLS Seminar, Los Alamos National Laboratory	06/2026
	<i>Adaptive front-capturing and front-tracking for multiscale fluid flows</i>	
	CNLS Seminar, Los Alamos National Laboratory	06/2025
	Multiphysics Models Conference, Los Alamos National Laboratory	06/2025
	<i>Computational Shock Development</i>	
	MultiMat Conference, Breckenridge	08/2024
	CASC Seminar, Lawrence Livermore National Laboratory	05/2024
	PDE & Applied Math Seminar, UC Davis	05/2024
	<i>PDE-based methods for multiscale computational fluid dynamics</i>	
	ANAG Seminar, Lawrence Berkeley Laboratory	09/2023
	Advanced Modeling and Simulation Seminar, NASA	09/2023
	Applied and Numerical PDEs Seminar, UC Berkeley	09/2023
	LANS Seminar, Argonne National Laboratory	08/2023
	<i>Efficient smooth adaptive meshing in multi-D with applications to gas dynamics</i>	
	Theoretical and Computational Astrophysics Network	05/2023

	T-3 seminar, Los Alamos National Laboratory	04/2023	
	EES-16 seminar, Los Alamos National Laboratory	04/2023	
	SOCAMS, Harvey Mudd College	05/2022	
	<i>A PDE-based multiscale framework for efficient gas dynamics simulations</i>		
	Student-Run Math and Applied Math Seminar, UC Davis	02/2023	
	<i>Adaptive mesh methods for an ignition-based fire model</i>		
	CSES symposium, Los Alamos National Laboratory	09/2022	
	<i>A multiscale model for unstable interfaces in compressible flows with vorticity</i>		
	Student-Run Math and Applied Math Seminar, UC Davis	05/2020	
	<i>PDE methods for the numerical simulation of compressible fluid flow</i>		
	Student-Run Math and Applied Math Seminar, UC Davis	04/2018	
CONFERENCES ATTENDED	<i>SIAM Conference on Analysis of Partial Differential Equations (PD22)</i>	03/2022	
	Virtual		
	<i>Convex Integration and Nonlinear Partial Differential Equations</i>	11/2021	
	ICMS, virtual		
	<i>Advances and Challenges in Hyperbolic Conservation Laws</i>	05/2021	
	ICERM, virtual		
	<i>Mathematical Analysis of Geophysical Models and Data Assimilation</i>	06/2020	
	Freie Universität Berlin, virtual		
	<i>AMS Spring Central and Western Joint Sectional Meeting</i>	03/2019	
	University of Hawaii at Manoa, Honolulu, HI, USA		
	<i>Conference on Mathematical Analysis of Incompressible Fluids</i>	06/2018	
	Mathematical Institute of University of Seville, Seville, Spain		
TEACHING EXPERIENCE	Spring 2023	MAT 119B	Ordinary Differential Equations
	Winter 2023	MAT 128B	Numerical Analysis
	Fall 2022	MAT 22B	Differential Equations
	Spring 2022	MAT 21C	Calculus
	Winter 2022	MAT 22B	Differential Equations
	Fall 2021	MAT 22B	Differential Equations
	Spring 2021	MAT 189	Advanced Problem Solving
	Winter 2021	MAT 16B & 17A	Calculus (Lead TA)
	Fall 2020	MAT 21A & 21C	Calculus (Lead TA)
	Spring 2020	MAT 127C	Real Analysis
	Winter 2020	MAT 127B	Real Analysis
SCIENTIFIC RESEARCH & OTHER WORK EXPERIENCE	Graduate Student Researcher, Los Alamos National Laboratory	2022–2023	
	Group: Methods & Algorithms XCP-4		
	<i>Adaptive mesh methods for an ignition-based fire model</i>		
	Visiting scholar, University of California, Davis	2017–2018	
	Advisor: Prof. Steve Shkoller		
	<i>PDE methods for computational physics</i>		

	Undergraduate research student, University of Oxford Advisor: Prof. Ian Hewitt <i>Modelling of ice sheets</i>	2015
	Summer school student, Perm State University <i>Fluid Dynamics, Perturbation Theory, & Lie Group Analysis</i>	2015
	Undergraduate research student, University of Oxford Advisors: Prof. Steve Shkoller & Prof. Ian Hewitt <i>Computational methods for gas dynamics</i>	2014
PROFESSIONAL ACTIVITIES & SERVICE	Reviewer for Journal of Computational Physics	2022–present
	UC Davis student-run Analysis and PDE seminar, organizer	2020–2022
	American Mathematical Society, member	2018–present
	Society for Industrial and Applied Mathematics, member	2021–present
	Graduate Student Assembly, representative	2019–2020
GRADUATE COURSEWORK	Real, complex, functional, harmonic analysis; linear & nonlinear PDEs; dynamical systems; calculus of variations; fluid mechanics; numerical methods for ODEs & PDEs; classical mechanics, quantum theory, relativity; algebra.	
RELEVANT SKILLS	Computing:	Fortran, C++, MATLAB, Python, OpenMP, MPI, Kokkos
	Languages:	English (fluent), Tamil (native)
	Other:	First Aid, CPR, & AED certifications from American Heart Association